vive la resistance

function

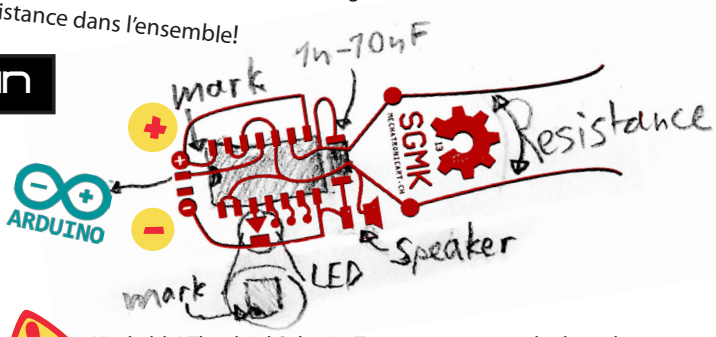
The Vive La Resistance aka NanosSchmano measures the air's resistance between the two antenna electrodes. Our manipulation of this resistance is audible via the oscillator.

The NAND-Schmitt-Trigger is very commonly used in digital technology. This element is accompanied by a condenser and a resistor, a so-called RC-circuit. By feeding back the signal we build an oscillator: A sound emerges! The assembly plan only shows the condenser, because we ourselves build the resistor via our breath or our fingers!

Be a part of the resistance dans l'ensemble!

assembly plan

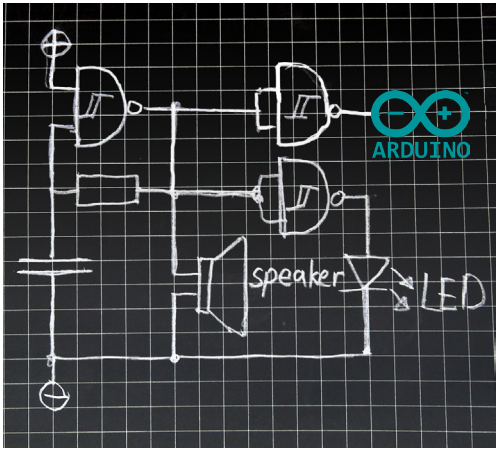
- Before soldering: polish the copper surface with sand paper for ideal response of the electrodes.
- The notch of the Schmitt-Trigger-Chip should be oriented towards the edge of the board.
- condenser: 1 bis 10nF.
-> have FUN!!



Hackable! The third Schmitt-Trigger-output on the board is accessible and can be connected to an Arduino, to measure e.g. the conductibility of the air via the frequency.
>> http://arduino.cc

advanced

The NAND-gate sends out the inverted incoming signal: With 9V at the input, 0V are output and vice versa. Via the loopback of the output signal the input signal gets inverted, however, with a delay. The big resistance causes the condenser to charge or discharge slowly. At a certain voltage the chip inverts the output signal, which in turn changes the charging of the condenser: The circuit oscillates!



schematics